

Exercise 1 of Monte Carlo Simulation

Mean balance	311528.11	=AVERAGE(I19:I118)
Sigma (=Risk)	6956.06	=STDEV.S(I19:I118)
Mean return	11528.11	=G7-G3

Randomizing		
Short-term	Intermediate	Longterm
m=2, sigma=0.3	m=5, sigma=2.5	m=6.25, sigma=5.5

Return	Investing 300000		
	Short-term	Intermediate	Longterm
	45	35	20
	135000	105000	60000
=E16/100*\$F\$14	=F16/100*\$F\$14	=G16/100*\$F\$14	
1475.678978	-2685.414033	-6225.949069	
2764.82648	5670.171631	4278.215724	
2349.357324	2977.315986	892.911526	
3053.672317	7542.320573	6631.774433	
2786.963576	5813.652804	4584.592099	
2679.558675	5117.509928	3583.441051	
2544.263584	4240.597305	2481.036611	
3209.845222	8554.552365	7904.294401	
3075.128434	7681.387998	6806.602055	
2968.856655	6992.589433	5940.683858	
2320.087681	2787.605338	654.4181398	
3135.611662	8073.40892	7299.428351	
2924.714609	6706.483574	5581.007921	
2713.742965	5339.074774	3861.979711	
2492.257364	3903.519953	2057.288222	
1820.951179	-447.5386542	-3412.620022	
2160.473509	1753.069042	-646.1417759	
2559.633715	4340.218253	2606.274711	
2275.548281	2498.924046	291.5045151	
2306.947881	2702.43997	547.35105058	
3621.077026	11219.94369	11255.07206	
2644.696629	4891.552222	3299.3799336	
2122.68968	2153.595161	-142.6232264	
1647.23951	-1573.44762	-1828.04843	
1740.445618	-969.3339545	-4068.591257	
2574.023587	4433.48621	2723.255522	
2732.177729	5458.559356	4012.188909	
2772.653359	5720.901398	4341.9903232	
2804.55836	5927.028567	4601.121622	
2810.135261	5963.839654	4647.398424	
2307.493032	2705.973357	551.7950778	
2640.419582	6354.571368	5138.604009	
2649.91424	4925.370077	3341.893811	
2546.2637	4253.561015	2497.333848	
2060.049674	1102.173811	-1464.410066	
2810.682254	5967.384978	4651.855401	
3017.297292	7306.556525	6335.385346	
3044.64141	7483.786915	6558.189265	
2720.193153	5380.881546	3914.536266	
2489.886141	3888.150913	2037.961148	
3167.810571	8282.105553	7561.7898383	
2944.130092	6832.324761	5739.208115	
3390.514935	9725.559763	9376.417988	
3285.072394	9042.135885	8517.2565454	
2700.016118	5509.36373	8076.05726	
2266.701547	2441.5841	219.4200116	
2661.447134	5000.12031	343.8655337	
2407.828465	3356.295607	1369.343048	
3141.626742	8112.395547	7348.440111	
2372.838419	3129.508271	1084.238969	
3012.268457	7273.962224	6294.409655	
3108.883602	7900.171496	7081.644166	
3804.063631	12405.96798	12746.07403	
4088.043165	12426.56707	15059.98131	
2814.707808	5993.476534	4684.656214	
2589.446667	4533.45062	2849.195061	
2447.155778	3611.194855	1689.781815	
2484.443838	3852.87673	1993.616464	
3102.995602	7862.008529	7033.66886	
2893.916538	47.42200492	-2790.383767	
2571.875673	4419.564546	2706.024001	
2863.468157	1772.478796	-621.7409424	
2196.259354	6457.236551	-5267.668801	
2057.117645	1083.169919	-1488.300673	
1738.382884	-982.7035266	-4085.398719	
3265.776622	8917.0707	8360.031738	
2458.983347	3687.855027	1786.160605	
2455.34916	3664.300109	1756.548798	
2789			

Ending balance
=F\$14+SUM(E19:G19)
292564.3159
312713.2139
306219.5848
317227.7673
313059.2085
311380.5097
309265.8975
319668.692
317563.1185
315902.1299
305762.1112
318508.4489
315212.2061
311914.7975
308453.0595
297960.7925
303267.4008
309506.127
305065.9768
305556.741
326096.0928
310835.6288
304233.2409
295245.7435
296702.5204
309731.0353
312202.926
312835.5451
313332.606
313421.3733
305565.2615
314363.595
310917.1781
309297.1586
301697.8134
313429.9226
316659.2392
317086.6176
312015.6115
308415.9982
319011.706
315515.6629
322492.4927
320844.4648
312325.4371
304927.7057
311097.4333
307133.4671
318602.4624
306586.5857
316580.6403
318090.6993
328956.1056
333394.6006
313492.8406
309972.0924
307748.1385
308330.937
317998.672
299154.3548
309697.4642
303314.206
314611.1647
301651.9869
296670.2806
320542.8791
307932.999
307876.198
313106.3304
314859.9794
317933.0698
315512.1799

	A	B	C	D	E	F	G	H	I	J
91	1.988671675	4.905597292	6.042314043		2684.706761	5150.877157	3625.388426		311460.9723	
92	1.753210659	2.943422159	1.725528749		2366.83439	3090.593267	1035.317249		306492.7449	
93	2.196472001	6.637266678	9.851986691		2965.237202	6969.130012	5911.192015		315845.5592	
94	1.976256731	4.802139428	5.814706741		2667.946587	5042.246399	3488.824044		311199.017	
95	1.968328439	4.736070322	5.669354708		2657.243392	4972.873838	3401.612825		311031.7301	
96	2.491018	9.091816663	15.25199666		3362.874299	9546.407496	9151.197996		322060.4798	
97	2.197126496	6.642720804	9.863985768		2966.12077	6974.856844	5918.391461		315859.3691	
98	1.629280865	1.910673872	-0.546517482		2199.529167	2006.207565	-327.9104893		303877.8262	
99	2.075265234	5.627210284	7.629862624		2801.608066	5908.570798	4577.917574		313288.0964	
100	1.912471935	4.27059946	4.645318812		2581.837113	4484.129433	2787.191287		309853.1578	
101	2.188408649	6.570072072	9.704158559		2954.351676	6898.575676	5822.495136		315675.4225	
102	2.082987867	5.69156556	7.771444233		2812.033621	5976.143838	4662.86654		313451.044	
103	2.054750444	5.456253701	7.253758143		2773.9131	5729.066387	4352.254886		312855.2344	
104	1.893535573	4.112796445	4.298152179		2556.273024	4318.436267	2578.891307		309453.6006	
105	1.691063977	2.425533138	0.586172904		2282.936368	2546.809795	351.7037423		305181.4499	
106	1.773479885	3.112332378	2.097131231		2394.197845	3267.948997	1258.278739		306920.4256	
107	1.943381908	4.52818257	5.212001655		2623.565576	4754.591699	3127.200993		310505.3583	
108	2.255586656	7.129888799	10.93575536		3045.041985	7486.383239	6561.453214		317092.8784	
109	2.01286719	5.107226583	6.485898483		2717.370706	5362.587912	3891.53909		311971.4977	
110	2.697793439	10.81494533	19.04287972		3642.021143	11355.69259	11425.72783		326423.4416	
111	2.203811624	6.698430196	9.986546432		2975.145692	7033.351706	5991.927859		316000.4253	
112	1.880800033	4.00666694	4.064667267		2539.080044	4207.000287	2438.80036		309184.8807	
113	1.712539875	2.604498958	0.979897708		2311.928831	2734.723906	587.9386246		305634.5914	
114	2.121537937	6.012816142	8.478195513		2864.076215	6313.456949	5086.917308		314264.4505	
115	1.993906613	4.949221774	6.138287903		2691.773927	5196.682863	3682.972742		311571.4295	
116	1.542283149	1.185692908	-2.141475603		2082.082251	1244.977553	-1284.885362		302042.1744	
117	2.157112254	6.309268782	9.130391321		2912.101543	6624.732221	5478.234793		315015.0686	
118	2.003614559	5.030121328	6.316266921		2704.879655	5281.627394	3789.760153		311776.2672	
119					=A118*F\$17/100	=B118*F\$17/100	=C118*G\$17/100		=F\$14+SUM(E118:G118)	